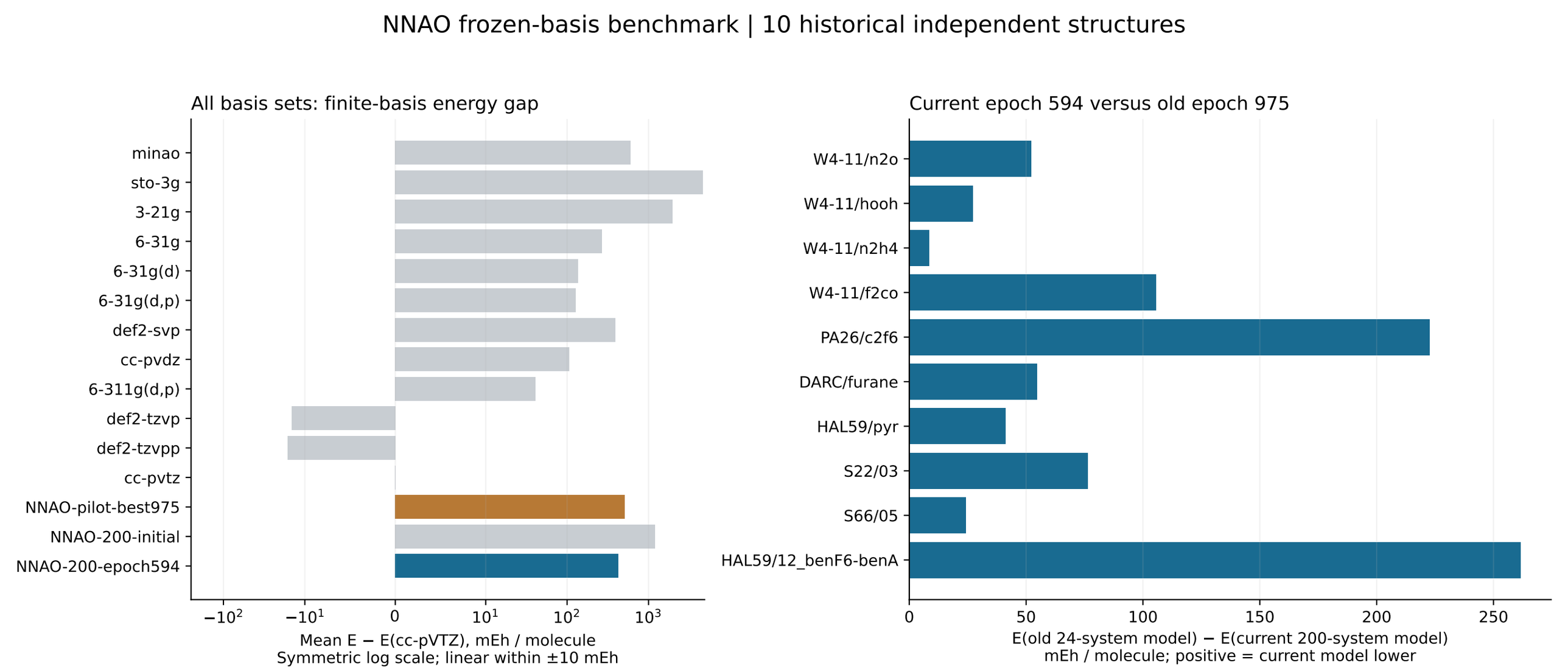
NNAO frozen-basis benchmark | 10 historical independent structures
All basis sets: finite-basis energy gap
minao
sto-3g
3-21g
6-31g
6-31g(d)
6-31g(d,p)
def2-svp
cc-pvdz
6-311g(d,p)
def2-tzvp
def2-tzvpp
cc-pvtz
NNAO-pilot-best975
NNAO-200-initial
NNAO-200-epoch594
−10²
−10¹
0
10¹
10²
10³
Mean E − E(cc-pVTZ), mEh / molecule
Symmetric log scale; linear within ±10 mEh
Current epoch 594 versus old epoch 975
W4-11/n2o
W4-11/hooh
W4-11/n2h4
W4-11/f2co
PA26/c2f6
DARC/furane
HAL59/pyr
S22/03
S66/05
HAL59/12_benF6-benA
0
50
100
150
200
250
E(old 24-system model) − E(current 200-system model)
mEh / molecule; positive = current model lower
All-electron DF-RHF | spherical AO | def2-universal-jkfit | cc-pVTZ is a finite-basis reference, not CBS